

Class Test 2

Date : 10 October 2018
Time : 6:00 – 7:30 pm

Marks available: 51
Full marks: 50

No calculators may be used.

1. Read the following theorem and the outline of its proof:

Theorem. Let $(f_k)_{k=0}^{\infty}$ be a sequence of complex-valued functions, each analytic on a simply connected region D containing a piecewise smooth simple closed curve $\mathcal{C}$. Suppose that $(f_k)_{k=0}^{\infty}$ converges uniformly on $\mathcal{C}$. Then $(f_k)_{k=0}^{\infty}$ converges on the interior of $\mathcal{C}$ to an analytic function f . Moreover, the sequence of derivatives $(f_k')_{k=0}^{\infty}$ converges on the interior of $\mathcal{C}$ to f' .

Proof. Start by defining $f(w) = \lim_{k \rightarrow \infty} f_k(w)$ for all $w \in \mathcal{C}$. We can extend this function to an analytic function on the interior of $\mathcal{C}$ by defining

$$f(z) = \frac{1}{2\pi i} \int_{\mathcal{C}} \frac{f(w)}{w - z} dw \quad (1)$$

for all z inside $\mathcal{C}$. For each k ,

$$f_k(z) - f(z) = \frac{1}{2\pi i} \int_{\mathcal{C}} \frac{f_k(w) - f(w)}{w - z} dw \quad (2)$$

for each z inside $\mathcal{C}$. Thus

$$|f_k(z) - f(z)| \leq \frac{|\mathcal{C}|}{2\pi d} \sup_{w \in \mathcal{C}} |f_k(w) - f(w)|, \quad (3)$$

where d is the distance from z to $\mathcal{C}$. Then $\sup_{w \in \mathcal{C}} |f_k(w) - f(w)| \rightarrow 0$ as $k \rightarrow \infty$, so $(f_k(z))_{k=0}^{\infty}$ converges to $f(z)$ for all z inside $\mathcal{C}$. Since

$$|f_k'(z) - f'(z)| \leq \frac{|\mathcal{C}|}{2\pi d^2} \sup_{w \in \mathcal{C}} |f_k(w) - f(w)| \quad (4)$$

for each k , it follows that $f_k'(z) \rightarrow f'(z)$ for all z inside $\mathcal{C}$. $\square$

Now answer the following questions:

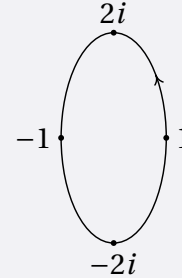
- Define what is meant by a function being *analytic* on the region D .
- Why can we define $f(w) = \lim_{k \rightarrow \infty} f_k(w)$ if $w \in \mathcal{C}$?
- Explain why equation (2) holds.
- Just after inequality (3), d is defined to be the distance from z to $\mathcal{C}$. How is this distance defined? Which property of the curve $\mathcal{C}$ is used here?

(e) Where exactly have we used the fact that $(f_k)_{k=0}^\infty$ converges *uniformly* on $\mathcal{C}$?

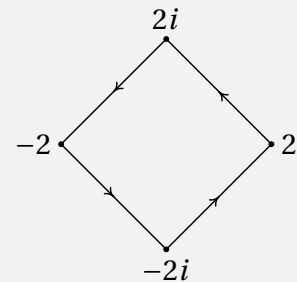
(f) Explain carefully how to obtain inequality (4).

[1, 1, 3, 2, 2, 4]

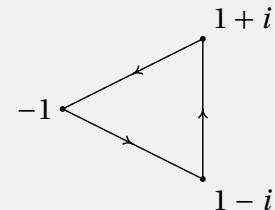
2. Find (a) $\int_\gamma \frac{e^{\pi z}}{(z-3)(z^2+1)} dz$, where γ traces out the curve



(b) $\int_\gamma \frac{8z^7}{z^8-1} dz$, where γ traces out the curve



(c) $\int_\gamma \left(\frac{z - \frac{\pi}{2}}{\sin z} \right)^2 dz$, where γ traces out the curve



[6, 4, 6]

3. Does there exist a function f that is analytic on the open disc $D_1 = \{z \in \mathbb{C} : |z| < 1\}$ and that satisfies

$$f\left(\frac{1}{n}\right) = \frac{(-1)^n}{n}$$

for all integers $n \geq 2$? (A “yes” or “no” answer won’t earn you any marks: you need to give a proper reason.) [4]

4.(a) Let f be analytic on the open disc $D_1 = \{z \in \mathbb{C} : |z| < 1\}$. Show that if $|f(z)| \leq 1 - |z|$ for all $z \in D_1$, then $f(z) = 0$ on D_1 .

(b) Can an analytic function $g: D_1 \rightarrow \mathbb{C}$ satisfy $|g(z)| \geq \frac{1}{1-|z|}$ for all $z \in D_1$? (“Yes” or “No” won’t earn you any marks.) [5, 3]

5. Let $p(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$, where $a_0, a_1, a_2, \dots, a_n \in \mathbb{C}$ and $a_n \neq 0$.

(a) Show that if $n > 0$, then there exists an $R > 0$ such that

$$|p(z)| \geq \frac{1}{2} |a_n| R^n$$

for all $z \in \mathbb{C}$ with $|z| \geq R$.

- (b) Explain carefully how part (a) can be used to prove the Fundamental Theorem of Algebra: *If the polynomial p has no zeros in $\mathbb{C}$, then p must be constant on $\mathbb{C}$.* Which theorem do you use to prove this result? [6, 4]

Class Test 2 solutions

1.(a) The function is differentiable at each point in D .

(b) Since $(f_k)_{k=0}^{\infty}$ converges on $\mathcal{C}$.

(c) Cauchy's integral formula says that

$$f_k(z) = \frac{1}{2\pi i} \int_{\mathcal{C}} \frac{f_k(w)}{w-z} dw$$

for all z inside $\mathcal{C}$. Thus

$$f_k(z) - f(z) = \frac{1}{2\pi i} \int_{\mathcal{C}} \frac{f_k(w)}{w-z} dw - \frac{1}{2\pi i} \int_{\mathcal{C}} \frac{f(w)}{w-z} dw.$$

(d) $d = \inf_{w \in \mathcal{C}} |w-z|$. This infimum exists because $\mathcal{C}$ is compact.

(e) We require uniform convergence to ensure that $(f_k)_{k=0}^{\infty}$ converges to a *continuous* function on $\mathcal{C}$ and to ensure that $\sup_{w \in \mathcal{C}} |f_k(w) - f(w)| \rightarrow 0$ as $k \rightarrow \infty$.

(f) Cauchy's formula for derivatives says that the function f defined by equation (1) is differentiable. This formula also says that

$$f'_k(z) = \frac{1}{2\pi i} \int_{\mathcal{C}} \frac{f_k(w)}{(w-z)^2} dw \text{ and } f'(z) = \frac{1}{2\pi i} \int_{\mathcal{C}} \frac{f(w)}{(w-z)^2} dw$$

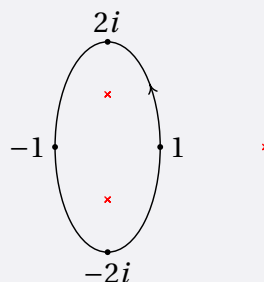
for all z inside $\mathcal{C}$. Thus

$$f'_k(z) - f'(z) = \frac{1}{2\pi i} \int_{\mathcal{C}} \frac{f_k(w) - f(w)}{(w-z)^2} dw$$

and so

$$|f'_k(z) - f'(z)| \leq \frac{1}{2\pi} \frac{\sup_{w \in \mathcal{C}} |f_k(w) - f(w)|}{\inf_{w \in \mathcal{C}} |w-z|^2} \cdot |\mathcal{C}|.$$

2.(a) There are simple poles at i , $-i$ and 3 , but only the first two of these are inside the curve traced out by γ :

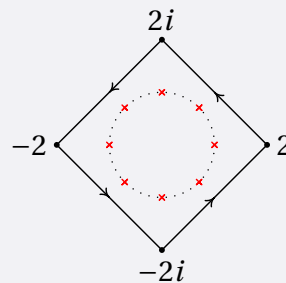


Thus

$$\begin{aligned}\int_{\gamma} \frac{e^{\pi z}}{(z-3)(z^2+1)} dz &= 2\pi i \left(\operatorname{Res}_{z=i} \frac{e^{\pi z}/(z-3)}{z^2+1} + \operatorname{Res}_{z=-i} \frac{e^{\pi z}/(z-3)}{z^2+1} \right) \\ &= 2\pi i \left(\frac{e^{\pi i}/(i-3)}{2i} + \frac{e^{-\pi i}/(-i-3)}{-2i} \right) \\ &= -\pi \left(\frac{1}{i-3} + \frac{1}{i+3} \right) \\ &= \frac{\pi}{5}i.\end{aligned}$$

(You could also apply Cauchy's integral formula.)

- (b) The integrand has simple poles at the points $z = \omega^k$, where $\omega = e^{i\pi/4}$ and $k = 0, 1, 2, \dots, 7$. These all lie inside the curve traced out by γ :



For each k , $\operatorname{Res}_{z=\omega^k} \frac{8z^7}{z^8-1} = \frac{8\omega^{7k}}{8\omega^{7k}} = 1$. Thus

$$\int_{\gamma} \frac{8z^7}{z^8-1} dz = 16\pi i.$$

Again, you could also apply Cauchy's integral formula. Alternatively, you could use the residue at infinity:

$$\begin{aligned}\int_{\gamma} \frac{8z^7}{z^8-1} dz &= -2\pi i \operatorname{Res}_{z=\infty} \frac{8z^7}{z^8-1} \\ &= 2\pi i \operatorname{Res}_{z=0} \frac{8z^{-7}}{z^2(z^{-8}-1)} \\ &= 2\pi i \operatorname{Res}_{z=0} \frac{8/(1-z^8)}{z} \\ &= 16\pi i.\end{aligned}$$

- (c) The integrand is undefined at all points where $\sin z = 0$, but we need only worry about $z = 0$. This is a double pole, so

$$\begin{aligned}\int_{\gamma} \left(\frac{z - \frac{\pi}{2}}{\sin z} \right)^2 dz &= 2\pi i \operatorname{Res}_{z=0} \left(\frac{z - \frac{\pi}{2}}{\sin z} \right)^2 \\ &= 2\pi i \operatorname{Res}_{z=0} \left(\frac{(z - \frac{\pi}{2})^2}{z^2} \left(\frac{z}{\sin z} \right)^2 \right) \\ &= 2\pi i(-\pi) = -2\pi^2 i.\end{aligned}$$

3. Note that 0 is a cluster point of $A = \{1/(2k) : k \in \mathbb{N}\}$. The function $g(z) = z$ satisfies $g(z) = f(z)$ for all $z \in A$, but $g(\frac{1}{3}) = \frac{1}{3} \neq f(\frac{1}{3})$. This contradicts the coincidence principle.

4.(a) Let $\gamma(t) = re^{it}$, where $0 \leq t \leq 2\pi$ and $0 < r < 1$. Cauchy's integral formula says that if $|z| < r$, then

$$|f(z)| \leq \frac{1}{2\pi} \frac{1-r}{r-|z|} \cdot 2\pi r = \frac{(1-r)r}{r-|z|}.$$

Letting $r \rightarrow 1$ shows that $f(z) = 0$.

(b) Notice that $g(z) \neq 0$ for all $z \in D_1$, so if g were analytic on D_1 , then so would be its reciprocal $f(z) = 1/g(z)$. But then $f(z)$ would be identically 0 by part (a), which is impossible.

5.(a) Using the reverse triangle inequality,

$$|p(z)| \geq \left| |a_n z^n| - |a_{n-1} z^{n-1}| - |a_{n-2} z^{n-2}| - \cdots - |a_1 z| - |a_0| \right|$$

for all $z \in \mathbb{C}$. Thus if $z \neq 0$,

$$\frac{|p(z)|}{|z|^n} \geq \left| |a_n| - (|a_{n-1}| |z|^{-1} + |a_{n-2}| |z|^{-2} + \cdots + |a_1| |z|^{1-n} + |a_0| |z|^{-n}) \right|.$$

The expression in brackets tends to 0 as $|z| \rightarrow \infty$. Thus it is possible to find an $R > 0$ such that it is smaller than $\frac{1}{2} |a_n|$ for all z with $|z| \geq R$. For such z ,

$$|p(z)| \geq \frac{1}{2} |a_n| |z|^n \geq \frac{1}{2} |a_n| R^n.$$

(b) It follows from part (a) that $1/p$ is bounded outside the disc $D_R = \{z \in \mathbb{C} : |z| < R\}$. Since it is continuous on $\overline{D_R}$, which is compact, it is bounded on $\overline{D_R}$ as well. Thus $1/p$ is bounded on $\mathbb{C}$. It is an entire function since it is a rational function with domain $\mathbb{C}$. Liouville's Theorem says that $1/p$ is constant. Thus p is constant on $\mathbb{C}$.